

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q003 **Written**

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q005 **Written**

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q007 **Written**

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ in QIV, find $\sin \theta$ and $\cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ in QIII, find $\cos \theta$ and $\tan \theta$.

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ in QIII, $\cos \theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Choose the correct option.

- A $\frac{7}{24}$
- B $\frac{24}{7}$
- C $-\frac{7}{24}$
- D $\frac{25}{7}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ **and** θ **in** QIII, **evaluate the exact value of** $\sin^6 \theta - \cos^6 \theta$.

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII, find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin \theta \cos \theta$ and $\sin^3 \theta - \cos^3 \theta$.

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Q023 Written**Given** $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{5}{16}$
B $\frac{11}{16}$
C $-\frac{11}{16}$
D $\frac{3}{16}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; (b) $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q034 Written

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 Written

Prove $\frac{\tan \theta}{1 + \tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$.

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Choose the correct option.**

- A Change both sides at once
- B Simplify one side
- C Substitute random values
- D Take a square root

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:

Choose the correct option.

A $\cos^2 \theta$

B $\tan^2 \theta$

C $\sec^2 \theta$

D $\sin^2 \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q040 MCQ

 $\tan \theta$ should be rewritten as:

Choose the correct option.

A $\sin \theta \cos \theta$

B $\sin \theta / \cos \theta$

C $\cos \theta / \sin \theta$

D $1 / \sin \theta$

Q041 MCQ**To add two trig fractions, first find a:****Choose the correct option.**

- A common numerator
- B common denominator
- C new quadrant
- D reference angle

Classified Practice

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Choose the correct option.**

- A LHS=0
- B LHS=RHS
- C RHS=1
- D undefined

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Choose the correct option.

- A $\frac{2\sqrt{5}}{5}$
B $-\frac{2\sqrt{5}}{5}$
C $\frac{\sqrt{5}}{5}$
D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q047 MCQ**If $s + c = 1$, then sc equals:****Choose the correct option.**

- A 0
- B $\frac{1}{2}$
- C $-\frac{1}{2}$
- D 1

Classified Practice

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{19}{27}$
B $\frac{5}{27}$
C $-\frac{19}{27}$
D $\frac{11}{27}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

- A $\frac{11}{16}$
B $-\frac{11}{16}$
C $\frac{7}{16}$
D $-\frac{7}{16}$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Choose the correct option.

- A $\frac{\sin \theta}{\cos \theta}$
B $\frac{\cos \theta}{\sin \theta}$
C $\sin \theta \cos \theta$
D $\sec \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Choose the correct option.**A $\sin^2 \theta$ B $\tan^2 \theta$ C $\cos^2 \theta$

D 1

Q053 MCQ**When proving an identity, which side should normally be transformed?****Choose the correct option.**

- A Both sides simultaneously
- B One side only
- C Neither side
- D Only the denominator

Classified Practice

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Choose the correct option.

- A A random angle
- B The Pythagorean identity
- C A logarithm
- D A derivative

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Choose the correct option.

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $-\frac{4}{5}$
- D $\frac{3}{4}$

Quiz MCQ 3 Concept Quiz · MCQ**Which identity is always valid where defined?****Choose the correct option.**

A $\tan \theta = \frac{\cos \theta}{\sin \theta}$

B $1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$

C $\sin \theta + \cos \theta = 1$

D $\sin \theta \cos \theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written**Given** $s + c = \frac{2}{3}$, **find** sc **and** $s^3 + c^3$.

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$ **Q018** $-\frac{55\sqrt{7}}{256}$ **Q032** LHS = $2 \csc x$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$ **Q019** $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$ **Q033** Identity proved

Q007 $\frac{3}{5}$, $\frac{4}{5}$ **Q020** $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$ **Q034** Identity proved

Q008 $-\frac{5}{13}$, $\frac{12}{13}$ **Q021** $-\frac{11}{16}$ **Q035** Identity proved

Q009 $-\frac{\sqrt{2}}{2}$, 1 **Q022** $\frac{\sqrt{11}}{3}$ **Q036** Identity proved

Q010 B **Q023** $\frac{3}{2}$ **Q037** Identity proved

Q011 A **Q024** A **Q038** B

Q012 B **Q025** B **Q039** A

Q013 C **Q026** B **Q040** B

Q014 A **Q027** A **Q041** B

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$